

CS:4980

Foundations of Embedded Systems

Asynchronous Model

Part II

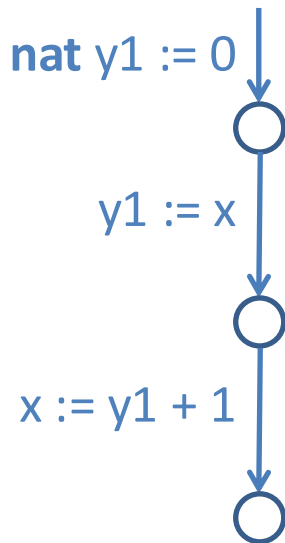
Copyright 2014-26, Rajeev Alur and Cesare Tinelli.

Created by Cesare Tinelli at the University of Iowa from notes originally developed by Rajeev Alur at the University of Pennsylvania. These notes are copyrighted materials and may not be used in other course settings outside of the University of Iowa in their current form or modified form without the express written permission of one of the copyright holders. During this course, students are prohibited from selling notes to or being paid for taking notes by any person or commercial firm without the express written permission of one of the copyright holders.

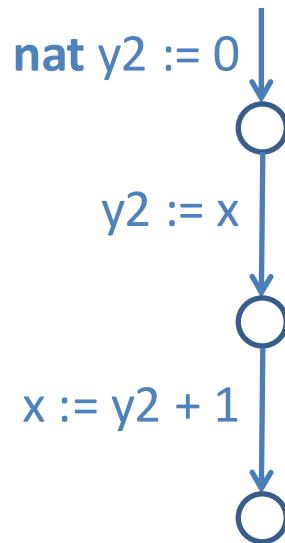
Shared Memory Programs

AtomicRegister nat $x := 0$

Process P1



Process P2



Declaration of **shared** variables
+ code for each process

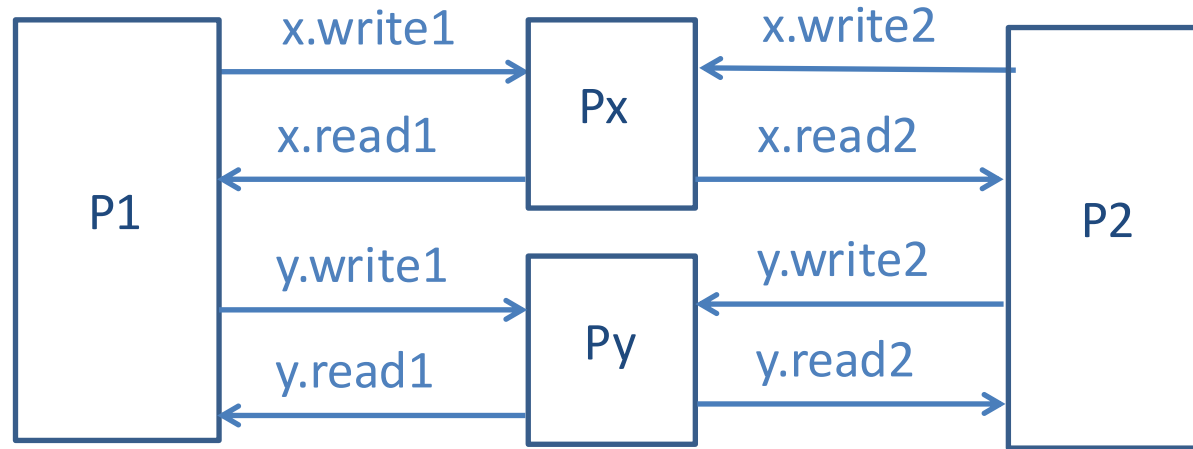
Key restriction: each process statement does one of the following

- changes local variables
- reads (single) shared var x
- writes shared var x

Execution model: execute one step of one of the processes

Can be formalized as **asynchronous processes**

Shared Memory Processes



Processes P_1 and P_2 communicate by reading/writing shared variables

Each shared variable s can be modeled as an asynchronous process P_s

For each such s

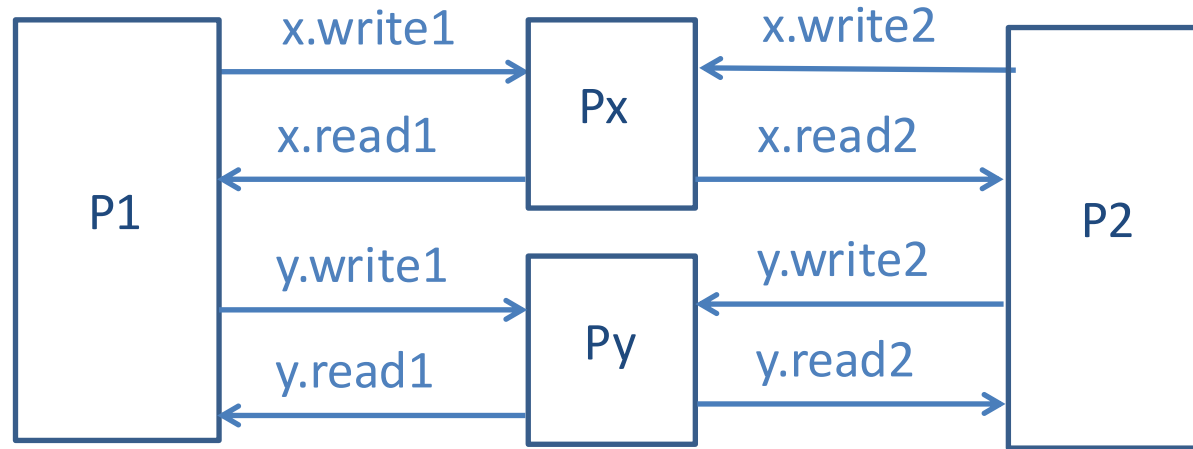
P_s has **output** channels $s.read1$, $s.read2$ and **input** channels $s.write1$, $s.write2$;

its **state** stores the value of s

Example above:

- To write x , P_1 synchronizes with P_x on $x.write1$ channel
- To read x , P_2 synchronizes with P_x on $x.read2$ channel

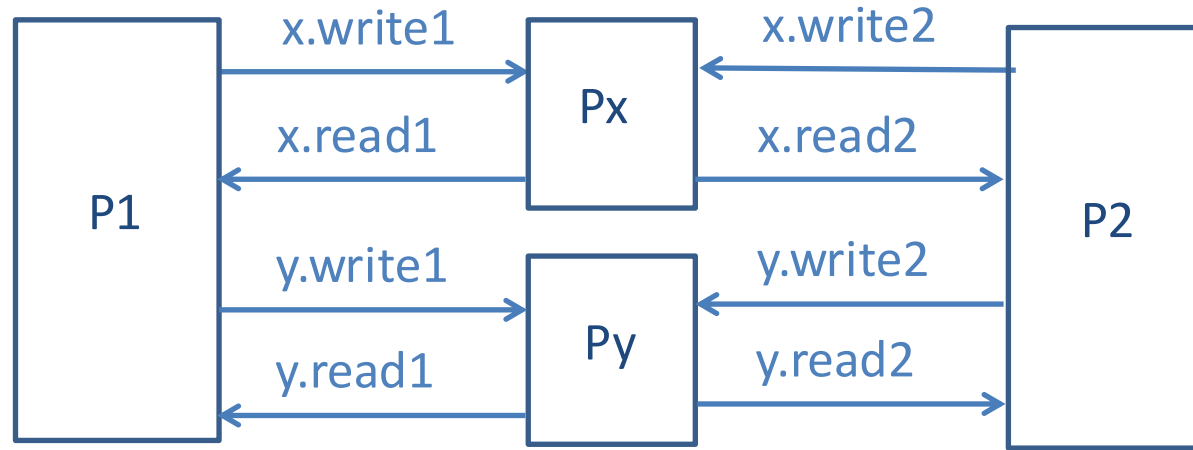
Atomic Registers



Note: By def. of asynchronous model, each step of above model is either

1. internal to **P1** or **P2**, or
2. involves exactly one synchronization
(read or write of one shared variable by one of the processes)

Atomic Registers



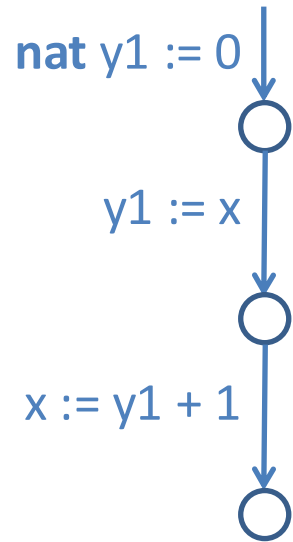
Atomic register: Basic primitives are **read** and **write**

- To **increment** such a register, a process first needs to **read** and **then write** back the incremented value
- But these **two are separate steps**, and register value can be changed in between by another process

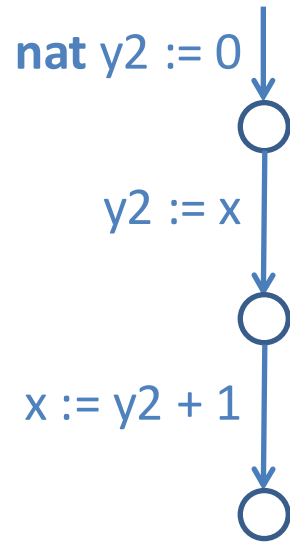
Shared Memory Programs

AtomicRegister nat x := 0

Process P1



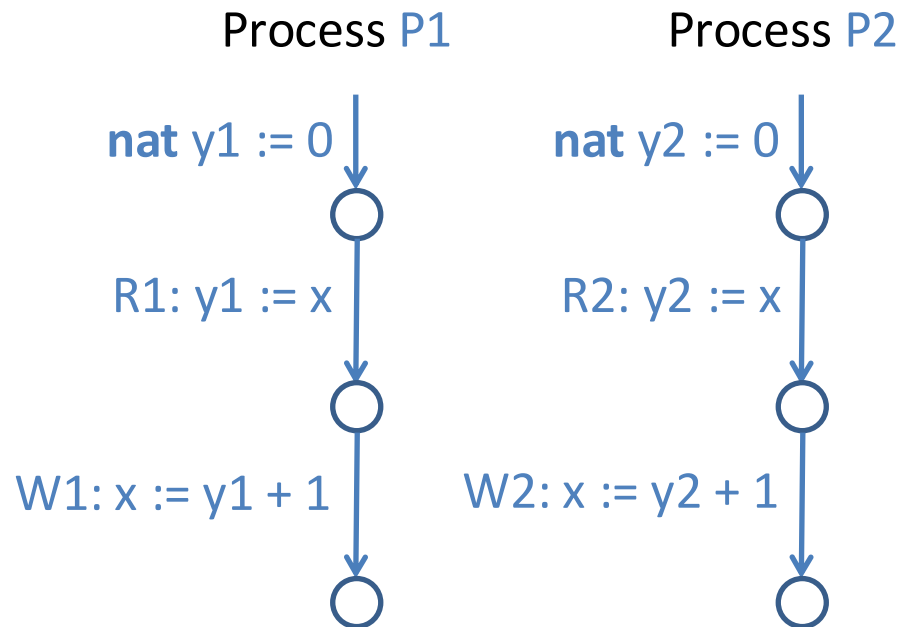
Process P2



Data Races

AtomicRegister nat x := 0

What are the possible values of **x** after all 4 read/write steps are executed?



x can be 1 or 2

Possible executions:

R1, R2, W1, W2

R1, W1, R2, W2

R1, R2, W2, W1

R2, R1, W1, W2

R2, W2, R1, W1

R2, R1, W2, W1

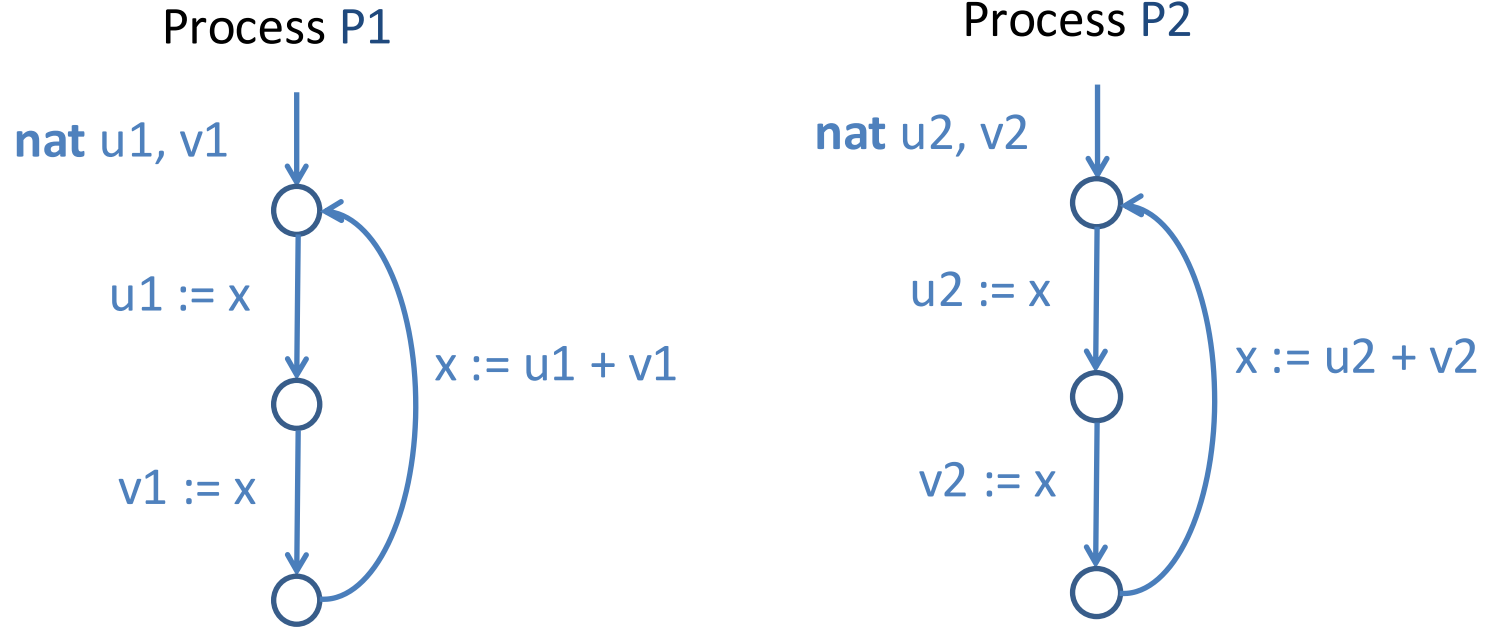
...

Data race: concurrent accesses to shared object where result depends on execution order

It should be avoided!

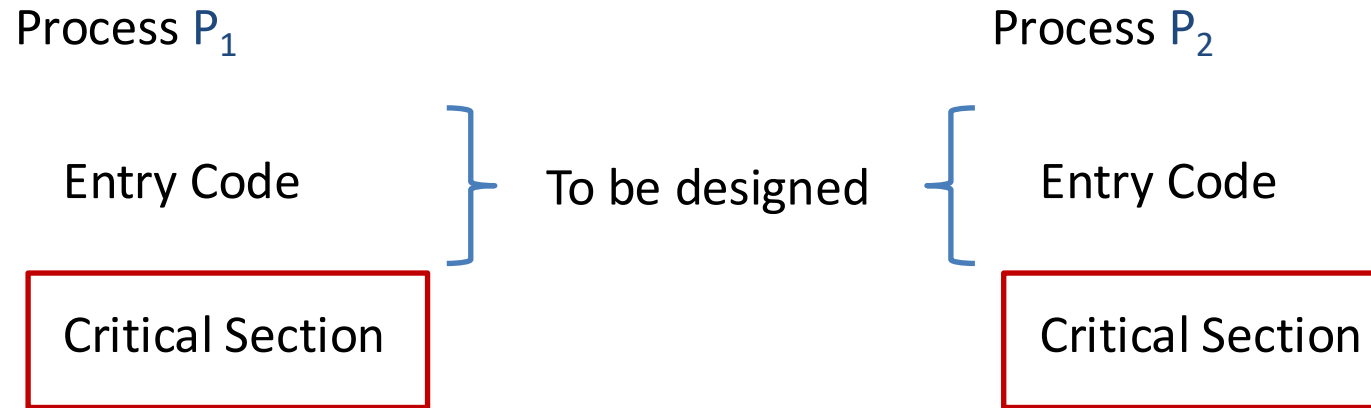
Exercise

AtomicRegister nat $x := 1$



What are the possible values for the shared register x ?

Mutual Exclusion Problem



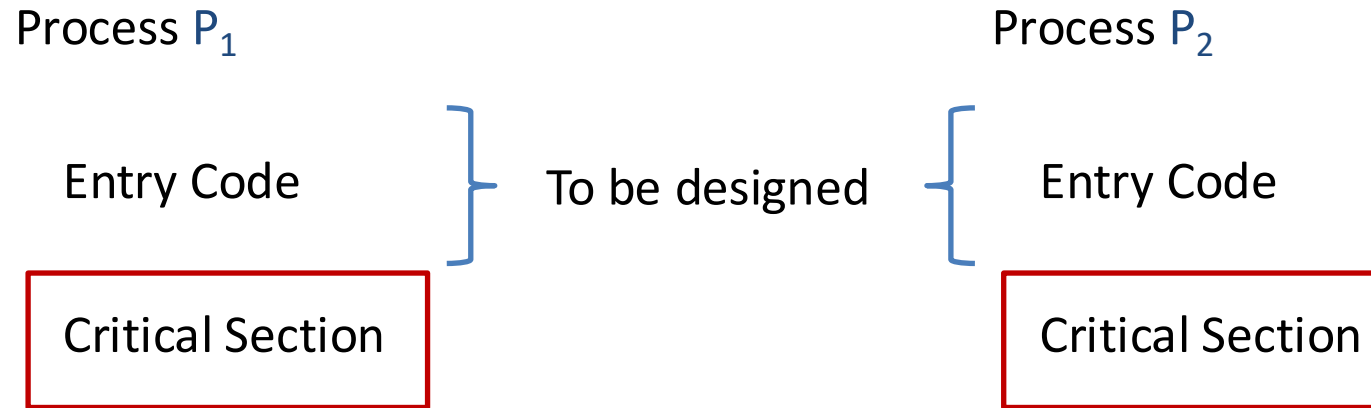
Critical Section: part of code that an asynchronous process should execute without interference from others

- Critical section can include code to update shared objects/database

Mutual Exclusion Problem: design code to be executed before entering critical section by each process

- Coordination using shared atomic registers
- No assumption about how long a process stays in critical section
- A process may want to enter critical section repeatedly

Mutual Exclusion Problem



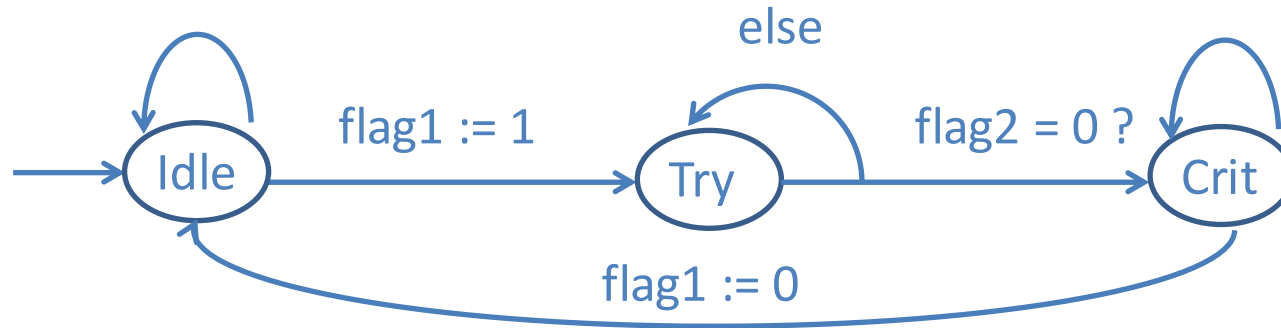
Safety requirement: both processes should not be in critical section simultaneously
(can be formalized using invariants)

Progress requirement: if any process is trying to enter, then some process should be able to enter (no deadlocks)

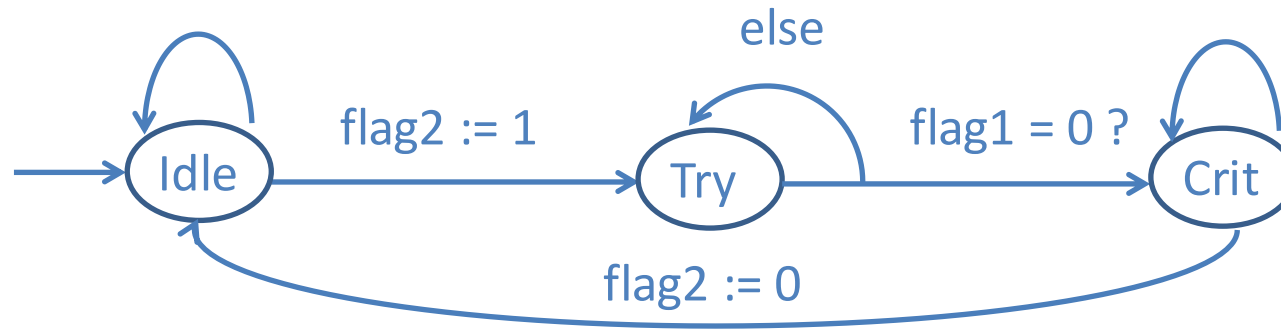
Mutual Exclusion: First Attempt

AtomicRegister bool flag1 := 0 ; flag2 := 0

Process P1



Process P2

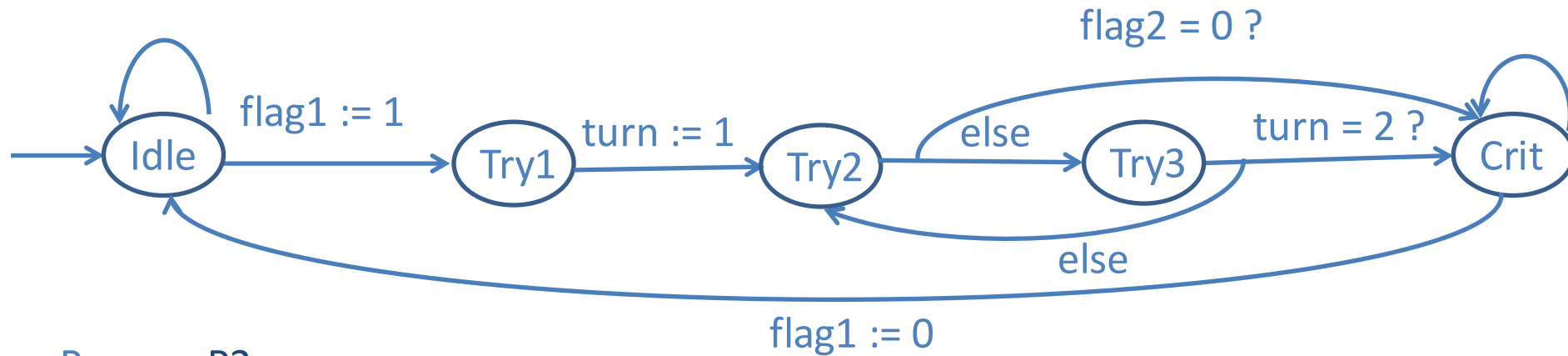


Is this correct?

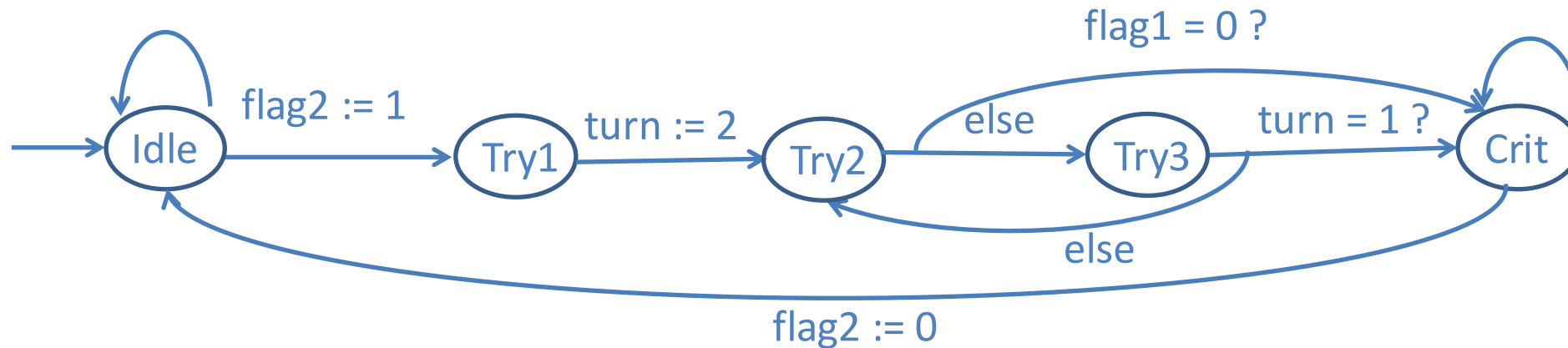
Peterson's Mutual Exclusion Protocol

AtomicRegister bool flag1 := 0 ; flag2 := 0 ; {1, 2} turn

Process P1



Process P2



Test&Set Register

Beyond atomic registers:

If we can do more than just read or write in one atomic step then we have stronger synchronization primitives

Test&Set Register: holds a Boolean value

- *Reset operation*: changes the value to 0
- *Test&Set operation*: returns the old value and changes value to 1
- If two processes are competing to execute *Test&Set* on a register with value 0, one will get back 0 and other will get back 1

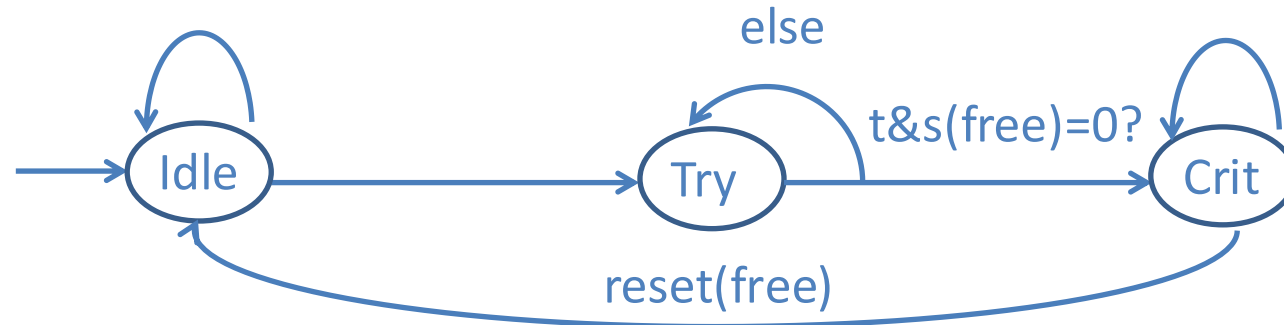
Modern processors support **strong atomic** operations

- **Example**: compare-and-swap; load-linked-store-conditional
- Implementation is expensive (compared to read/write operations)

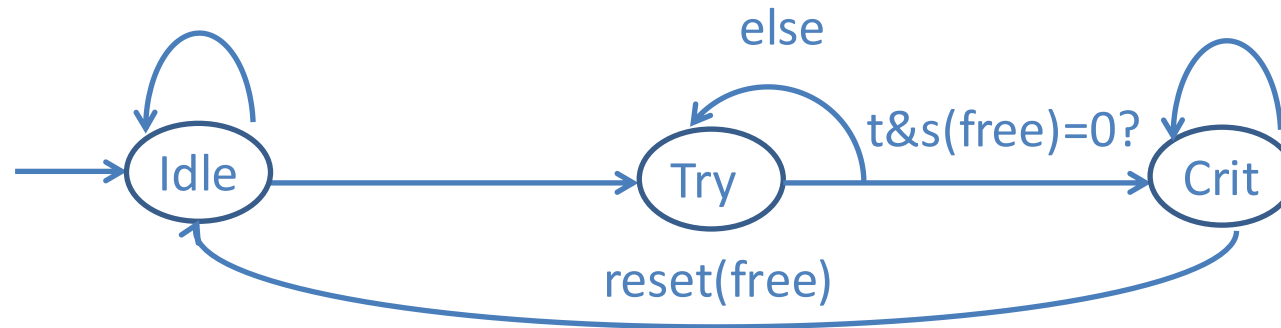
Mutual Exclusion using Test&Set Register

Test&SetRegister nat free := 0

Process P1



Process P2



Is this correct?

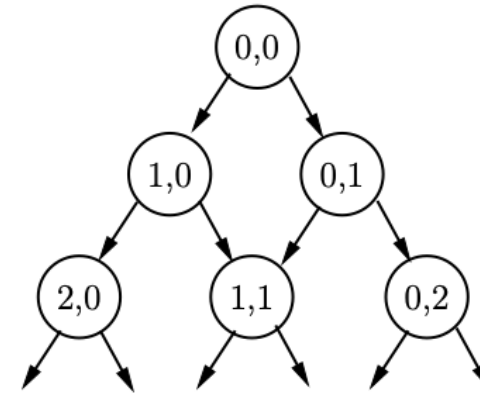
Another Look at Asynchronous Execution Model

Process P

```
nat x := 0 ; y := 0
```

```
Ax: x := x + 1
```

```
Ay: y := y + 1
```



Tasks A_x and A_y execute in an arbitrary order

Motivation: If we establish that all possible executions of this design satisfy some requirement R , then every implementation of P will satisfy R

Are the following realistic executions?

- $(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_x} (3,0) \dots \xrightarrow{A_x} (95,0) \xrightarrow{A_x} \dots$
- $(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_y} (2,1) \xrightarrow{A_y} (2,2) \dots \xrightarrow{A_y} (2,95) \dots$

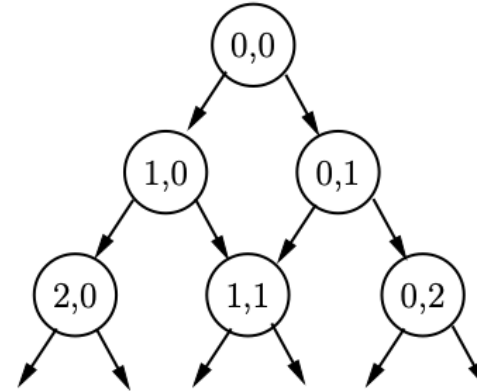
Does the system satisfy the following requirement:

In every execution, values of both x and y eventually exceed 10

Fairness Assumption

Process P

<code>nat x := 0 ; y := 0</code>
<code>A_x: x := x + 1</code>
<code>A_y: y := y + 1</code>



Fairness assumption for a task

- Assumption about the underlying platform/scheduler
- Informally, an infinite execution is *unfair* to a task if the task does not get a chance to execute

Unfair to A_y : $(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_x} (3,0) \dots \xrightarrow{A_x} (95,0) \dots$

Unfair to A_x : $(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_y} (2,1) \xrightarrow{A_y} (2,2) \dots (2,95) \dots$

Fairness assumptions restrict the set of *possible* executions to *realistic* ones *without* imposing *concrete bounds* on relative speeds

Formalizing Fairness

Process P1

<code>nat x := 0 ; y := 0</code>
<code>A_x: x := x + 1</code>
<code>A_y: y := y + 1</code>

Process P2

<code>nat x := 0 ; y := 0</code>
<code>B_x: x := x + 1</code>
<code>B_y: x = 0 → y := y + 1</code>

Definition 0. An infinite execution is *fair* to a task *A*, if the task *A* is executed repeatedly (i.e., infinitely often) during that execution

- Is this execution fair to task *A_y*? How about *B_y*?
 $(0,0) \rightarrow_{B_x} (1,0) \rightarrow_{B_x} (2,0) \rightarrow_{B_x} (3,0) \rightarrow_{B_x} \dots (105,0) \rightarrow_{B_x} \dots$
- After first step, the task *B_y* is not enabled, and so cannot be executed. This execution should not be considered unfair

Definition 1 (Weak fairness). An infinite execution is *fair* to a task *A* if, repeatedly, either *A* is executed or is disabled (*If enabled then eventually executed or disabled*)

Weak vs Strong Fairness

Process P3

```
nat x := 0 ; y := 0
```

```
Ax : x := x + 1
```

```
Ay : even(x) → y := y + 1
```

- Is this execution fair to task A_y?

$(0,0) \xrightarrow{A_x} (1,0) \xrightarrow{A_x} (2,0) \xrightarrow{A_x} (3,0) \dots \xrightarrow{A_x} (105,0) \xrightarrow{A_x} \dots$

- According to weak fairness, yes, because A_y is disabled infinitely often

Definition 2 (Strong fairness). An infinite execution is *fair* to a task A, if task A is either executed repeatedly or disabled continuously from a certain step onwards
(If repeatedly enabled then repeatedly executed)

- Above execution is weakly fair to task A_y, but not strongly fair

Fairness Assumption

Fairness assumptions for an asynchronous process **P**:

For each output and internal task, either

- no assumption,
- weak fairness assumption, or
- strong fairness assumption

Restricts the **set of** possible infinite **executions**

- If weak/strong fairness is assumed for a task **A**, then task scheduling should be such that executions are weakly/strong fair to **A**

Makes it **easier** for a process **P** to **meet** a **requirement** **R**

- Maybe not all executions satisfy **R**, but all **fair** executions do

Requirements under Fairness Assumptions

Process P1

nat $x := 0$; $y := 0$
$A_x: x := x + 1$
$A_y: y := y + 1$

Process P3

nat $x := 0$; $y := 0$
$B_x: x := x + 1$
$B_y: \text{even}(x) \rightarrow y := y + 1$

Under what fairness assumptions do **P1** and **P3** satisfy these requirements?

R1: eventually, $x + y > 10$

- **P1** and **P3** both satisfy this, without any fairness assumption

R2: eventually, $x > 10$

- **P1** with weak fairness for A_x , **P3** with weak fairness for B_x

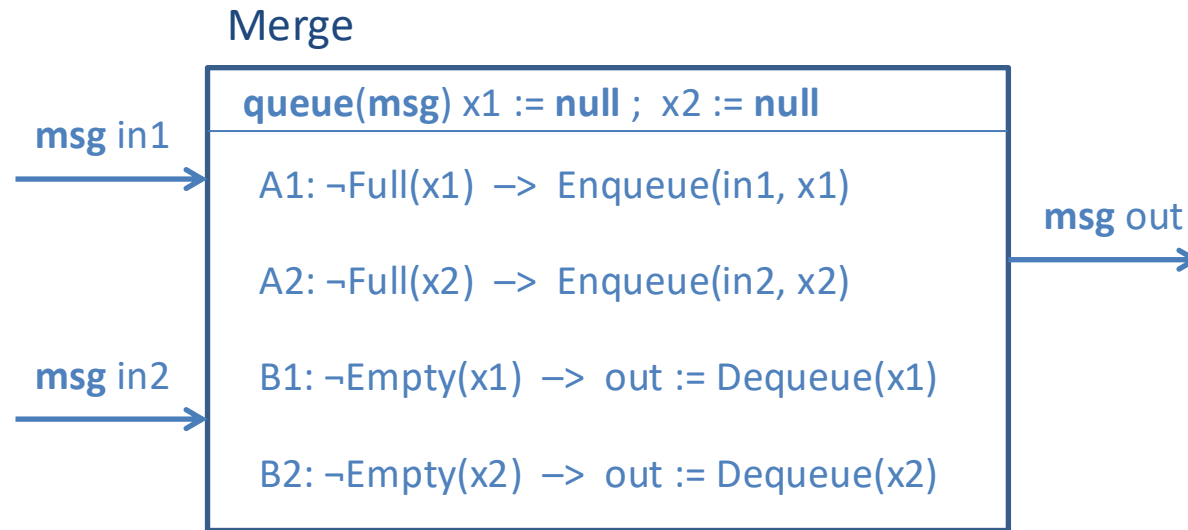
R3: eventually, $y > 10$

- **P1** with weak fairness for A_y , **P3** with strong fairness for B_y

R4: eventually, $x > y$

- neither, no matter what fairness assumption we make!

Asynchronous Merge

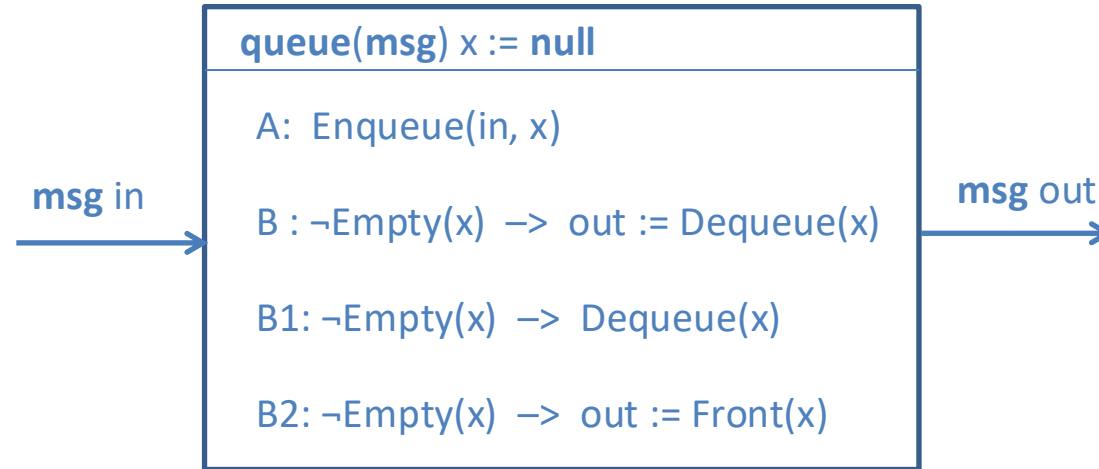


Requirement: whenever an input message is received, it is eventually output

Under which fairness assumptions does the requirement hold?

Weak fairness for tasks B1 and B2 suffices

Unreliable (Unbounded) FIFO



Tasks **A** and **B** model normal input/output behavior

Task **B1** models message loss

Task **B2** models message duplication

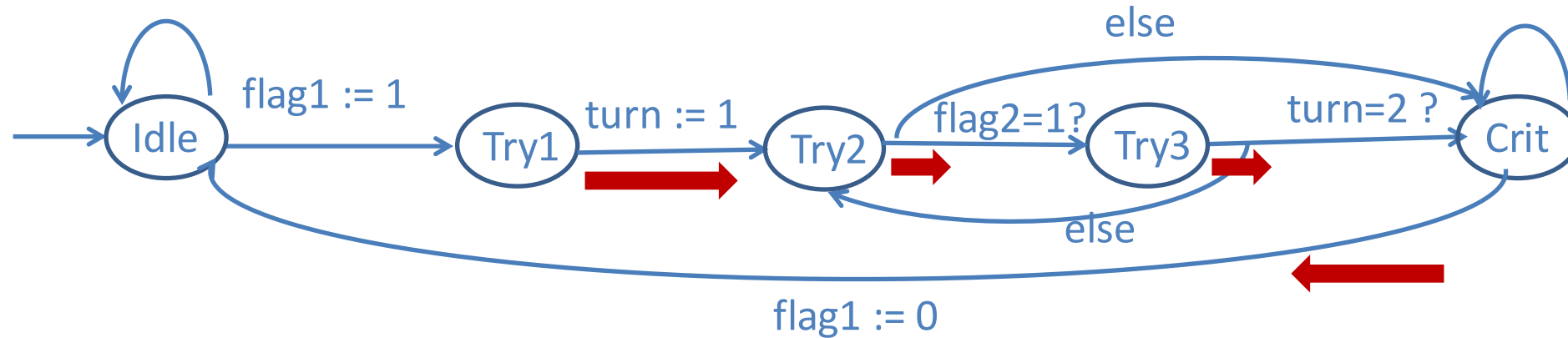
What are **natural** fairness assumptions for these tasks?

Strong fairness for **B**, no assumptions for **B1** and **B2**

Fairness Assumptions for Mutual Exclusion Protocol

AtomicRegister bool flag1 := 0 ; flag2 := 0 ; { 1, 2 } turn

Process P1



Requirement: if a process ever wants to enter critical section, it eventually will

What fairness assumptions should we make?

Weak fairness for highlighted steps/tasks

Fairness Summary

A fairness assumption is an **assumption** made **about** the underlying **platform** or **scheduler**

- The weaker the assumption, the better

It **restricts** the **set of** possible infinite **executions**, allowing the **satisfaction** of **more requirements**

- Does not affect the set of reachable states and safety properties
- Does not change underlying coordination

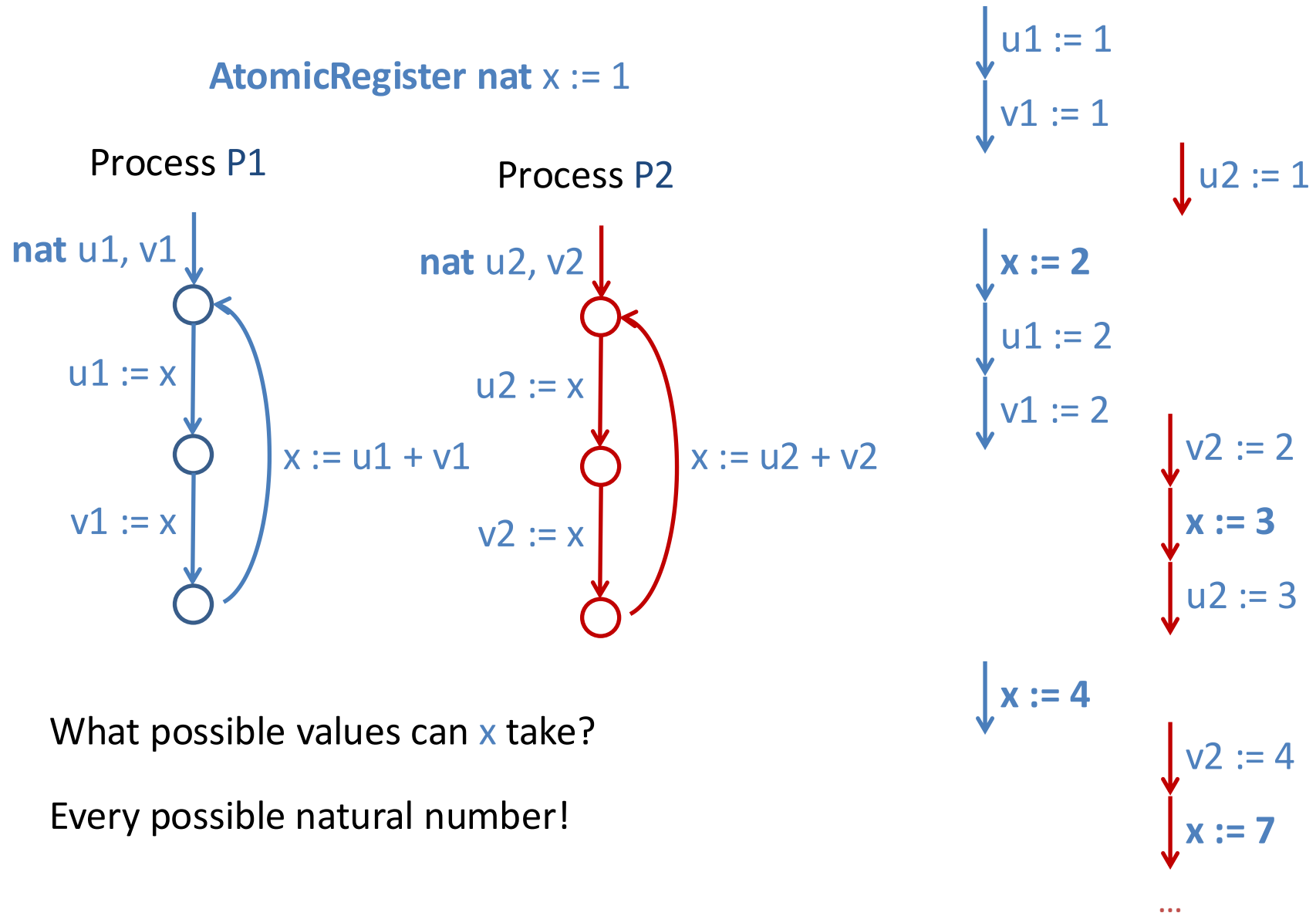
For each output and internal task, we can **assume weak** or **strong fairness**, as needed

- Strong fairness is needed if the task can switch between enabled and disabled due to the execution of other tasks

Key distinction:

- Fairness assumption for tasks (which ensures tasks get executed as expected) vs.
- *Fairness* requirements for protocols (which are about high-level goals of the problem being solved)

Exercise: Solution



Credits

Notes based on Chapter 4 of

Principles of Cyber-Physical Systems

by Rajeev Alur

MIT Press, 2015